Group No.:5

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Smart Automatic Door System Using IR Proximity Sensor and Servo Motor

Problem statement:

In many buildings such as offices, laboratories, and smart homes, doors are still operated manually. This traditional method can lead to inconvenience and inefficiency, especially in places where people frequently enter and exit.

Manual door systems create several problems:

- Users must physically open and close the door every time.
- It becomes inconvenient when the user's hands are occupied.
- There is no automated way to detect a person approaching the door.
- No record is maintained of when the door was opened or closed.
- Security monitoring becomes difficult without event logging.

Therefore, there is a need for an automated system that can detect when a person approaches a door and operate the door automatically.

The Smart Door Automation System using an IR Proximity Sensor addresses this problem by automatically detecting a user's presence near the door and controlling a servo motor to open or close the door.

With the advancement of embedded systems and smart sensors, automation has become an important part of modern infrastructure. This project focuses on designing a system that detects user approach using an IR proximity sensor, activates a servo motor to control the door mechanism, and logs the events of door opening and closing.

This system improves convenience, enhances automation, and allows monitoring of door usage through event logging.

Objectives

The main objectives of this project are:

- To design a system that detects user approach using an IR proximity sensor.
- To automate the opening and closing of a door using a servo motor.
- To reduce manual effort in operating doors.
- To record and log door operation events.
- To demonstrate the integration of sensors, actuators, and a microcontroller.

Methodology

Step 1: System Setup
The system is designed by integrating an IR proximity sensor, a servo motor, and a microcontroller. The IR sensor is placed near the door to continuously monitor the surrounding area for any approaching user.

Step 2: User Detection
The IR proximity sensor emits infrared light and detects the reflected signal when a person approaches the door. Once the reflected signal is detected, the sensor generates an output signal and sends it to the microcontroller.

Step 3: Signal Processing
The microcontroller receives the signal from the IR sensor and processes the input to determine whether a user has approached the door.

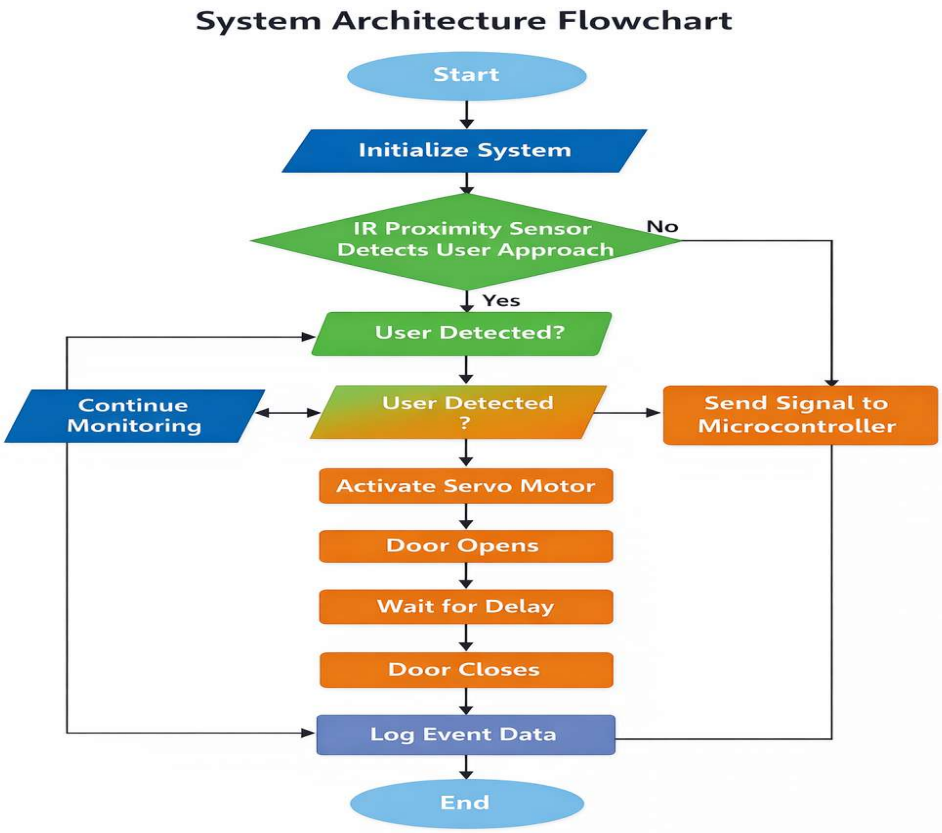
Step 4: Door Opening Mechanism
When the presence of a user is detected, the microcontroller sends a control signal to the servo motor. The servo motor rotates to a predefined angle which represents the door opening position.

Step 5: Door Closing Mechanism
After a fixed time delay, the microcontroller sends another signal to the servo motor to rotate back to its initial position. This action closes the door automatically.

Step 6: Event Logging
Each activity in the system is recorded, such as user detection, door opening, and door closing. These events are logged for monitoring and analysis purposes.

Step 7: Continuous Monitoring
The system continuously monitors the area using the IR sensor to detect the next approaching user and repeat the same process.

Flowchart



Technologies Used

1. IR Proximity Sensor
The IR proximity sensor is used to detect the presence of a user near the door. It

works by emitting infrared light and sensing the reflected signal when an object or person comes close to the sensor.

2. Microcontroller (Arduino / Embedded Controller)
The microcontroller acts as the main control unit of the system. It receives input signals from the IR sensor, processes the data, and sends control signals to the servo motor to perform the required action.

3. Servo Motor
The servo motor is used to control the opening and closing of the door. It rotates to a specific angle to open the door when a user is detected and returns to its original position to close the door after a certain delay.

4. Embedded Programming
Embedded programming is used to write the control logic for the system. The program enables communication between the sensor, microcontroller, and servo motor.

5. Event Logging Mechanism
The system includes an event logging mechanism that records activities such as user detection, door opening, and door closing. This helps in monitoring and maintaining records of system operations.

System Design

The system design of the Smart Automatic Door System is based on the integration of sensing, control, and actuation components to perform automated door operations.

1. Sensor Module

The sensor module consists of an IR proximity sensor which is placed near the entrance of the door. The sensor continuously monitors the surrounding area for the presence of a user. When a person approaches the door, the infrared rays emitted by the sensor are reflected back and detected by the receiver, generating a signal.

2. Control Unit

The control unit is the microcontroller, which acts as the brain of the system. It receives the signal from the IR sensor and processes the information. Based on the received input, the microcontroller decides whether to activate the door mechanism.

3. Actuation System

The servo motor is used as the actuation component of the system. When the microcontroller detects a user approaching the door, it sends a control signal to the servo motor. The servo motor rotates to a predefined angle to open the door and then returns to its original position to close the door after a delay.

4. Event Logging Module

The system includes an event logging mechanism that records important activities such as user detection, door opening, and door closing. These events can be stored in memory or transmitted to a monitoring system for further analysis.

5. Continuous Monitoring

The entire system works in a continuous loop where the IR sensor keeps detecting the presence of users. Whenever a user approaches, the system repeats the process of detection, door actuation, and event logging automatically.

Functional Components

1 IR Proximity Sensor

The IR proximity sensor detects when a user approaches the door.
It emits infrared light and senses the reflected signal from nearby objects or persons.
When detection occurs, the sensor sends a signal to the microcontroller.

2 Microcontroller (Control Unit)

The microcontroller acts as the brain of the system.
It performs two main actions:

1. Receive Sensor Input
   • Detects signals coming from the IR proximity sensor.

2. Control Door Operation
   • Sends commands to the servo motor to open or close the door.

3 Servo Motor (Door Actuator)

The servo motor controls the physical movement of the door.

1. Door Opening
   • The servo motor rotates to a specific angle to open the door when a user is detected.

2. Door Closing
   • After a delay, the servo motor returns to its initial position to close the door.

4 Event Logging System

The system records important events such as:
• User detected
• Door opened
• Door closed

This helps in monitoring and maintaining records of door operations.

Features of the System

• Automatic user detection
• Contactless door operation
• Quick response using sensors and actuators
• Event logging for monitoring
• Efficient and automated door control system

Advantages

• Provides **automatic and contactless door operation**, improving user convenience.
• Reduces manual effort and makes entry and exit easier for users.
• Improves hygiene and accessibility, especially in public places.

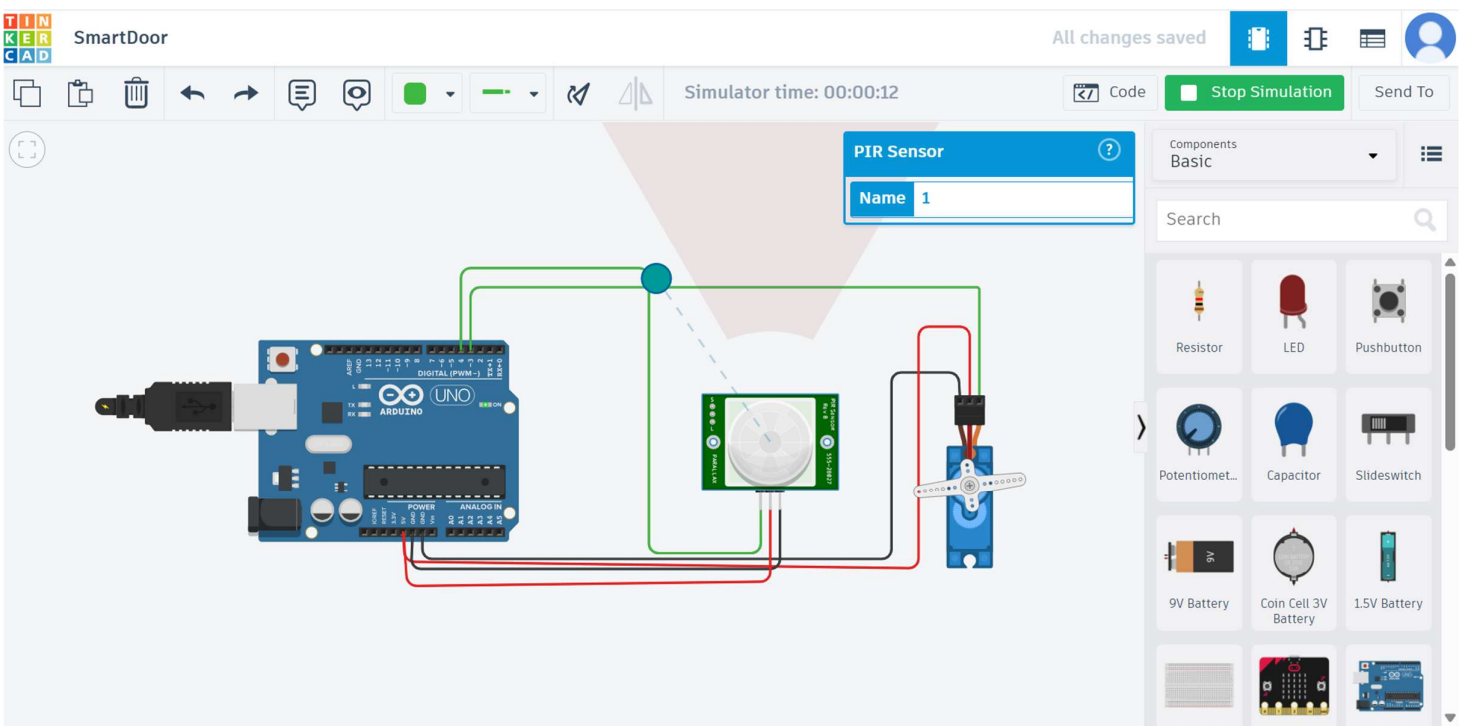
Limitations

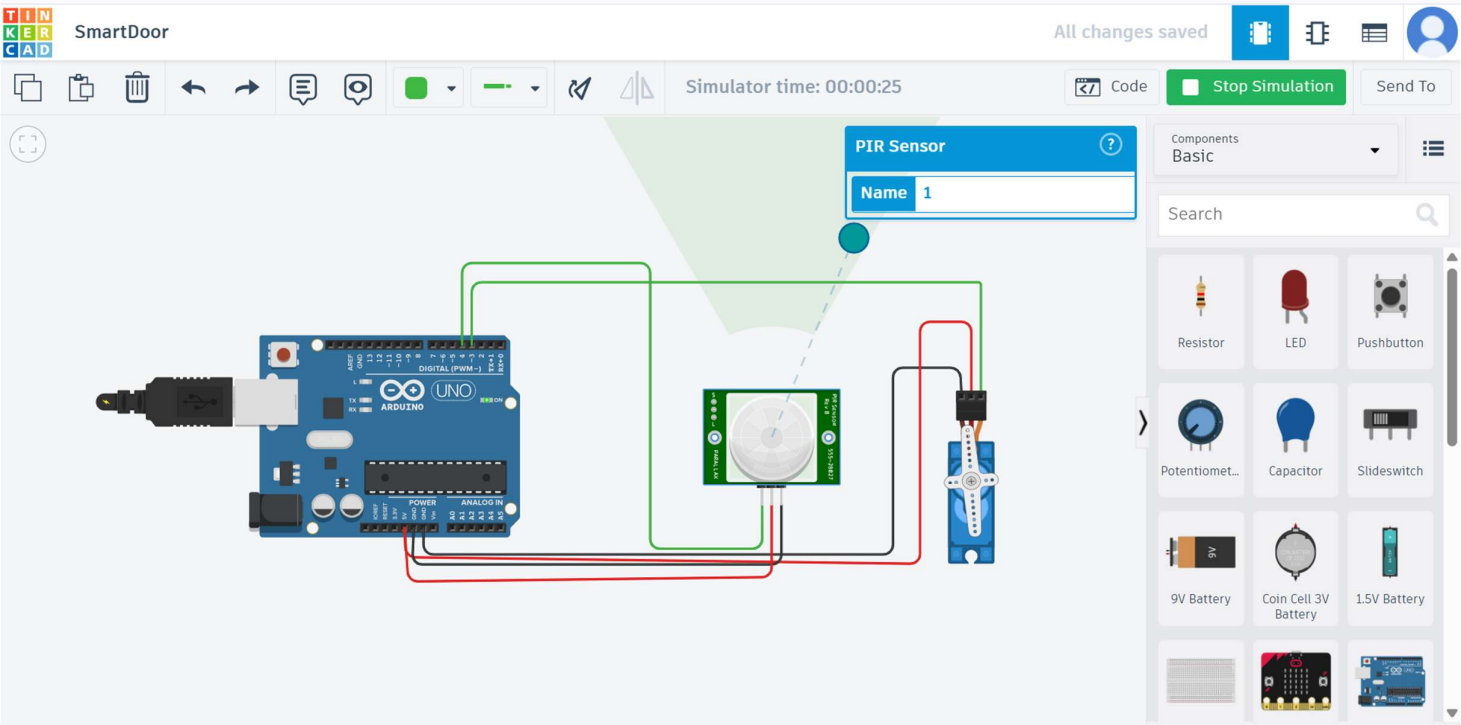
• IR sensors may be affected by environmental factors such as strong light or obstacles.
• The system works only within a limited detection range.
• Continuous power supply is required for proper operation.

Future Enhancements

• Integration with IoT systems for remote monitoring and control.
• Implementation of advanced security features like RFID, password systems, or facial recognition.
• Adding data storage and analytics to track door usage patterns.

OUTPUT SCREENSHOT





CODE:

```
#include<Servo.h>

Servo s;

int pir=4;

void setup()

{

  pinMode(pir,INPUT);

  s.attach(3);

  Serial.begin(9600);

}

void loop()

{
```

```
  int val=digitalRead(pir);
  Serial.println(val);
  if(val==HIGH)
  {
    s.write(180);
  }
    else
  {
    s.write(0);
  }
  delay(2000);
}
```

Conclusion

The Smart Automatic Door System using an IR proximity sensor and servo motor provides an efficient solution for automated door control. The system detects the presence of a user using the IR sensor and automatically opens or closes the door using a servo motor. It also logs important events such as user detection and door operations for monitoring purposes. This system improves convenience, reduces manual effort, and supports contactless operation. Such automation systems can be widely used in smart homes, offices, hospitals, and public buildings to enhance accessibility and efficiency.